
A-level
ENVIRONMENTAL SCIENCE
7447/2

Paper 2

Mark scheme

June 2023

Version: Final v1.0



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Qu	Part	Marking guidance	Comments	Total marks	AO
01				1 1 1 1 1	AO1 1a
		Term	Definition		
		Species	A group of organisms that resemble each other more than other organisms and interbreed to produce fertile offspring.		
		Biome	(Large geographical) region with particular climatic features and a unique community/vegetation.		
		Community	The populations of all the species living in a particular area.		
		Population	All the individuals of a species living in a particular area.		
		Ecological niche	The role an organism plays in its habitat (its use of resources and its inter-relationships with other species).		
		Total =		5	

Qu	Part	Marking guidance	Comments	Total marks	AO
02	1	- photosynthesis produced oxygen/O₂ (O₂) formed ozone/O₃ (O₃) <u>absorbs</u> UV radiation		1 1 1	AO1 1b

Qu	Part	Marking guidance	Comments	Total marks	AO
02	2	Two functions of CO₂ Two linked explanations how CO₂ aids survival Functions - used in photosynthesis - greenhouse gas/absorbs IR radiation Linked explanations - produces carbohydrates/sugars/proteins/lipids for growth/food chains - produces oxygen for (aerobic) respiration - maintains optimum/warmer temperatures		2 2	AO1 1a = 2 AO1 1b = 2

Qu	Part	Marking guidance	Comments	Total marks	AO
02	3	2013–2020 increased growth (wider rings) (increased growth) indicates higher temperatures/more rainfall [A: converse 1964–2012] [R carbon dioxide]		1 1	AO3 1b = 1 AO2 = 1

Qu	Part	Marking guidance	Comments	Total marks	AO
02	4	One from: - named example of other variable affecting growth rate eg disease, nutrients - multiple climate factors affect growth - no trees/limited age in some areas - only shows local climate - trees evolved relatively recently - trees decompose (so limited data records)		1	AO1 1b
			Total =	10	

Qu	Part	Marking guidance	Comments	Total marks	AO
03	1	number of different species in an area [R density]		1	AO1 1a

Qu	Part	Marking guidance	Comments	Total marks	AO
03	2	- measures sound <u>frequency</u> is unique to species		1 1	AO1 1b

Qu	Part	Marking guidance	Comments	Total marks	AO
03	3	Three from: - females fly over greater distance /longer flight path - females fly slower - females fly higher/over a greater height range - females fly over/through to wind turbines whereas the male fly around them - females north-south whereas males east-west/different direction [A converse]		3	AO3 1a

Qu	Part	Marking guidance	Comments	Total marks	AO
03	4	Three from: - designation of protected areas/named designation e.g. SSSI - reduce/prevent named threats along flight paths e.g wind turbines - manage food/water sources at feeding sites/on flight path - put up bat boxes/protect trees/buildings at roosting sites/on flight path - create/maintain biological corridors to connect feeding habitats/avoid threats - timings of flights can inform street lighting regimes - identify potential release sites		3	AO2

Qu	Part	Marking guidance	Comments	Total marks	AO
03	5	One from: • disturbs/may cause harm • tag may influence behaviour/survival • short battery life • fall off • some species not suitable to attach transmitter [A expensive]		1	AO2
			Total =	10	

Qu	Part	Marking guidance	Comments	Total marks	AO
04	1	Transect: • an environmental gradient is present/ (predicted) change over distance		1	AO2
		Random sampling: • reduces bias		1	

Qu	Part	Marking guidance	Comments	Total marks	AO
04	2	One from: 0.5m × 0.5m/0.25m² 1m × 1m/1m² 2m × 2m/4m² [R answers without units]		1	AO3 1b

Qu	Part	Marking guidance	Comments	Total marks	AO
04	3	One mark for the feature One mark for related explanation - grid - smaller areas allow totalling of each square or (quadrat with) points - vegetation touched (by point) is recorded/totalled		1 1	AO3 1b

Qu	Part	Marking guidance	Comments	Total marks	AO
04	4	One mark for correct distribution of named vegetation type and related to rate of retreat One mark for correct explanation • Early sere/named early sere taxa e.g. lichens, mosses, grasses covering a larger area/distance would suggest a fast rate of retreat • time for colonisation/succession [A converse]		1 1	AO2

Qu	Part	Marking guidance	Comments	Total marks	AO
04	5	Three explanations of how species change conditions: e.g. • weather minerals from rocks to provide soil nutrients • create shade to protect from UV/modifies temperatures • new species is a food source • new species provides breeding sites/shelter from predators • decomposition/addition of dead organic matter increases nutrients • addition of dead organic matter increases moisture retention • addition of dead organic matter increased soil depth • roots bind soil/substrate reducing erosion • species changes named abiotic factor e.g. intercept wind reducing wind speeds		3	AO1 1b
			Total =	10	

Qu	Part	Marking guidance	Comments	Total marks	AO
05	1	For level 3 both similarities and differences must be discussed/ ecosystems compared. Similar threats to both coral reef ecosystems • destructive fishing methods: damage the seabed destroying feeding/breeding grounds • demersal trawling scrapes away seabed • demersal longlining snags corals • shellfish traps damage on impact • ghost fishing from discarded fishing gear kills species • overfishing removes species above the MSY resulting in population decline • ocean temperature increase: beyond range of tolerance/breakdown in interspecies relationships/causing reduced oxygen • ocean acidification: reduction in pH and calcium carbonate saturation reduces growth rate Deep-water coral reefs only • hydrocarbon exploration/production causes physical impact from oil platforms • pipelines causes impacts from mud discharges from rock drilling • potential threat from oil spilled from mechanical fault at site of extraction • telecommunication/electricity cables • burial in the seabed disturbs sediment into water smothering coral • potential treat from future mineral exploration with the increase in AUVs/ROVs eg metals from deep sea nodules • changes in ocean currents that deliver nutrients Tropical coral reefs only • tourism: damage from boat anchors/ touching/ disturbance/ souvenir collection • pollution from surface run off: sewage/fertiliser causing eutrophication • increase turbidity/sedimentation from soil erosion due to development/ agriculture/ destruction of mangroves • introduced species • coral bleaching		9	AO1 1a = 2 AO1 1b = 2 AO2 = 3 AO3 1c = 2

Examiners are reminded that AO1, AO2 and AO3 are regarded as interdependent. When deciding on a mark all should be considered together using the best fit approach. In doing so, examiners should bear in mind the relative weightings of the assessment objectives. More weight should therefore be given to AO1 than AO2 and AO3.

Level	Marks	Descriptor
3	7–9	A comprehensive response to the question, with the focus sustained. A conclusion is presented in a logical and coherent way, fully supported by relevant judgements. A wide range of knowledge and understanding of natural processes/systems is applied. The answer clearly identifies relationships between environmental issues. Relevant environmental terminology is used consistently and accurately throughout, with no more than minor omissions and errors.
2	4–6	A response to the question which is focussed in parts but lacking appropriate depth. A conclusion may be present, supported by some judgements, but it is likely not all will be relevant. A range of knowledge and understanding of natural processes/systems is shown. There is an attempt to apply this to the question, but there may be a few inconsistencies, errors and/or omissions. The answer attempts to identify relationships between environmental issues, with some success. Environmental terminology is used, but not always consistently.
1	1–3	A response to the question which is unbalanced and lacking focus. It is likely to consist of fragmented points that are unrelated. A conclusion may be stated, but it is not supported by any judgments and is likely to be irrelevant. A limited range of knowledge and understanding of natural processes/systems is shown. There is an attempt to apply this to the question, but there are fundamental errors and/or omissions. The answer may attempt to identify relationship between environmental issues, but is rarely successful. Limited environmental terminology is used, and a lack of understanding is evident.
	0	Nothing written worthy of credit.

Qu	Part	Marking guidance	Comments	Total marks	AO
05	2	One from: • Marine Protected Area (MPA) • Marine Conservation Zone (MCZ) • Marine Nature Reserve (MNR) • Special Area of Conservation (SAC) • No Take Zones (NTZ) [A abbreviations]		1	AO1 1a

Qu	Part	Marking guidance	Comments	Total marks	AO
05	3	Two marks for difference in conditions - no light for photosynthesis - lower temperatures		2	AO2
		Two marks for reason conditions reduce rate of recovery - lower energy input - reduced growth rate/ rate of biochemical reactions [A converse]		2	

Qu	Part	Marking guidance	Comments	Total marks	AO
05	4	more interspecies relationships/ stable food web		1	AO2
		Total =		15	

Qu	Part	Marking guidance	Comments	Total marks	AO
06	1	80 and 60 $80 / 60 = 1.33$ or 1.3 or 1.33 : 1 ecf [R 1: 1.33] Award 2 marks for correct answer	Allow ecf if wrong data is divided correctly e.g. if data for pigs used 50 and 30 = no marks 50/30 = 1.67 1 mark ecf	1 1	AO3 1a AO2

Qu	Part	Marking guidance	Comments	Total marks	AO
06	2	B (Dairy Cattle)		1	AO3 1b

Qu	Part	Marking guidance	Comments	Total marks	AO
06	3	- energy used for artificial lighting - energy used for heating - energy used for machinery - material used for barn - production/growth of chicken feed		2	AO2
		Total =		5	

Qu	Part	Marking guidance	Comments	Total marks	AO
07	1	transfer of DNA/genetic material from one species to another		1	AO1 1a

Qu	Part	Marking guidance	Comments	Total marks	AO
07	2	Four from: - same <u>species</u> of caterpillar used in both fields - caterpillars fed on the same diet before the field trial - caterpillars should be same age - caterpillars were protected from predators in both fields - same number of caterpillars placed on each weed plant in each field/ same number weed plants used - same <u>species</u> of weeds used in both fields/use weed species not toxic to caterpillars - same positioning of weeds used eg on all sides of field edge/side of prevailing wind/distance from crop - same pest control methods / pesticides used in both field - investigation carried out when the pollen was being released from corn/corn at same stage of growth - fields should be far enough apart to prevent contamination of Bt pollen to non-GM corn weeds - same density of corn in both fields - same (area for same) climatic/weather conditions		4	AO3 1a

Qu	Part	Marking guidance	Comments	Total marks	AO
07	3	Three from: - mean mass/growth rate higher in Field B - mean mass significantly different on day 5/9/11 - difference between the mean mass increases over time. - variation in mass/standard deviation in Field A is higher on day 5 (onward) 9/11 - variation in mass in Field A increases with time [A converse]		3	AO3 1a

Qu	Part	Marking guidance	Comments	Total marks	AO
07	4	Up to two marks for reason One mark for explanation - wind may blow pollen to one side of the field more - caterpillars on one side will eat / exposed to more toxic Bt pollen than the other side - a natural genetic variability of caterpillar - different sensitivity to toxin / different growth rate - greater variability of weeds - some more nutritious / toxic - variation in competition - some caterpillars have access to less food [A converse]		1 1	AO3 1c

Qu	Part	Marking guidance	Comments	Total marks	AO
07	5	no (significant) difference in the percentage (%) survival of caterpillars in Bt and non-GM corn / Field A and B [R no correlation, mean mass]		1	AO2

Qu	Part	Marking guidance	Comments	Total marks	AO
07	6	13		1	AO2

Qu	Part	Marking guidance	Comments	Total marks	AO
07	7	- use of smaller U value/23 - greater than > critical value, so no significant difference /accept the null [A ecf converse of answer if answer to 07.6 is > 23]		1 1	AO3 1c

Qu	Part	Marking guidance	Comments	Total marks	AO
07	8	0.05 or 5% probability that chance caused the difference [A 95% confidence/significance]		1	AO1a

		Total =	15	
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Qu	Part	Marking guidance	Comments	Total marks	AO
08	1	73.3 (%)	$16100 + 1900 = 18\,000$ $67\,300 - 18\,000 = 49\,300$ $(49\,300 / 67\,300) \times 100 = 73.25$	1	AO2

Qu	Part	Marking guidance	Comments	Total marks	AO
08	2	Two changes to farming as a result of subsidy e.g. • beetle banks • buffer strips • organic farming/reduced pesticides • hedgerow maintenance/planting/removal • skylark plots • field margins/pollinating flower margins • drainage wetland • set aside/rewilding • renewable energy • use of machinery • use of pesticides Two impacts on the environmental linked to change e.g. • increase/decrease in death of non-target species • decrease/increase in nutrient leaching/eutrophication • increase/decrease in habitats • increase/decrease of food sources • increase/decrease biodiversity • decrease/increase fossil fuel use/CO₂ emissions Allow one mark for named government subsidy		2 2	AO2
			Total =	5	

Qu	Part	Marking guidance	Comments	Total marks	AO
09	1	• $1467 / 399 = 3.67669$ • 3.68 ecf Two marks for correct answer with no working	ecf for inverse division 0.272 ecf for wrong selected data from table, to correct number of significant figures	2	AO3 1a

Qu	Part	Marking guidance	Comments	Total marks	AO
09	2	Two marks for reasons. Two marks for linked explanations. Area: • largest total area • habitat destruction / named negative ecological impact e.g. increase competition of food Productivity: • highest productivity / high number of shrimps • highest inputs of named energy subsidies e.g. pesticides/ antibiotics/ food/machinery • highest output of waste product		4	AO3 1b = 2 AO2 = 2

Qu	Part	Marking guidance	Comments	Total marks	AO
09	3	Two marks for how changes in water quality occur: • food/faeces/high DOM • high rates of decomposition (of DOM) • pesticide/antibiotic residue/ hormones • high amounts bacteria/parasites Two marks for change in water quality: • increased turbidity • deoxygenation • high concentration of ammonium/nitrates • toxicity/antibiotic resistant bacteria • increased pathogens		2 2	AO2
			Total =	10	

Qu	Part	Marking guidance	Comments	Total marks	AO
10	1	One mark for correct region: - South America One mark for correct rate of deforestation for South America: 12 000 ecf One mark for answer in standard form: 1.2×10^4 ecf (A 1.2000×10^4) Max two mark for one ecf Max one mark for two ecf	Africa: $75 - 35 = 40$ $0.4 \times 20\,000 = 8\,000$ South America: $95 - 65 = 30$ $0.3 \times 40\,000 = 12\,000$ (Sub)tropical Asia: $70 - 30 = 40$ $0.4 \times 15\,000 = 6\,000$ ecf for correct rate of deforestation of incorrect region ecf for correct standard form of incorrect region	1 1 1	AO3 1a

Qu	Part	Marking guidance	Comments	Total marks	AO
10	2	Three marks for loss of forest ecosystem service: - reduced carbon sequestration / oxygen production - reduced interception - reduced evapotranspiration - reduced root binding of soil - loss of wildlife habitat - loss of amenity use - loss of named resource e.g. medicines / food - loss of genetic resource Three marks for linked impact on humans: - named impact of climate change / ozone formation - named impact of flooding eg loss of crops, homes - named impact of reduced rainfall/drought e.g. crop failure - named impact of soil erosion eg loss of land that supports agricultural, reduced quality of water for named use - less control of disease / malnutrition - loss of potential characteristics in agricultural species (from breeding programmes) - reduced income from tourism / resource exploitation - named impact of amenity loss		3 3	AO2

Qu	Part	Marking guidance	Comments	Total marks	AO
10	3	One from: • International Tropical Timber Organisation / ITTO • Forest Stewardship Council / FSC • Forestry Commission/ Forestry England [A abbreviations]		1	AO1 1a
		Total =		10	

Qu	Part	Marking guidance	Comments	Total marks	AO
11	1	Indicative mark scheme: Students should name methods and discuss both advantages and disadvantages		25	AO1 = 10 AO2 = 10 AO3 = 5
Topic area / spec ref		Method	Advantages	Disadvantages	
3.5.1.1 Agroecosystems 3.5.1.3 Environmental impacts of agriculture 3.5.1.5 Strategies to increase the sustainability of agriculture		Nutrient application methods eg Organic manures Green manures Crop rotation Legumes Buffer strips Conservation of soil biota	Reduced energy used/GHG produced from manufacture of artificial fertilisers Reduced eutrophication risk DOM provides food for soil biota/increases aeration/drainage/water retention reducing the need for as much irrigation/reduces risk of erosion Makes use of waste product crop rotation also provides pest control	Smell of manure Potential pathogen risk Run off causing organic nutrient pollution/eutrophication Manure heavy to transport Green manures take up growing space May not be a market for legume crops Buffer strips take crop land	
		Alternative to pesticide use eg Maintenance of natural predator habitats Introduction of natural predators Sterile male technique Crop rotation Barrier/sacrificial crops Buffer strips Mulching/weeding	Reduced death of non-target species/human poisonings Maintenance of pest predator habitats/buffer strips also increase general biodiversity Crop rotation can reduce fertiliser input/encourages diversification Mulching also provides interception reducing risk of soil erosion/evaporation from soil and need for as much irrigation Government subsidies	Less effective as pesticides/ loss of yield from pests Reduces land for crops Makes use of large machinery more difficult No market	
3.2.5.1 How human activities affect soil fertility					
3.2.5.2 Causes of soil degradation and erosion					
3.2.5.3 Soil management strategies to increase sustainability					
3.5.1.2 Manipulation of food species					

	Soil management methods eg Direct drill/zero tillage Cover/long-term crops Multi-cropping/strip cropping Contour/across slope ploughing increase DOM Tied ridging/terracing Windbreaks Mulching Drainage	Reduced soil erosion by interception/root binding/less disturbance/wind velocity Reduced impact on aquatic habitats Reduced release of CO ₂ /N ₂ O/CH ₄	Labour intensive/no machinery access DOM bulky to transport/energy use
	Genetic manipulation to reduce the need for additional inputs Selective breeding Asexual reproduction/ Vegetative propagation/ cloning Transgenics/GM	Higher yields Increases productivity reducing need for more land Pest resistance reduces need for pesticides Drought tolerance can reduce irrigation/over-abstraction impacts Nitrogen fixation can reduce fertiliser application/run off/NO _x Named examples eg Direct seeded rice can reduce methane emissions	Reduction of gene pool Potential impacts of GM/genetic contamination/toxic pollen Public opposition Cost
	Reduced use water use Drip irrigation Selective breeding GM Hydroponics	Reduced abstraction impacts Reduced run off impacts	Cost of new materials GM impacts
		Total =	25

Qu	Part	Marking guidance	Comments	Total marks	AO
11	2	Indicative mark scheme: Students should name methods and discuss both advantages and disadvantages		25	AO1 = 10 AO2 = 10 AO3 = 5
Topic area / spec ref		Method	Advantages	Disadvantages	
3.5.2.2 Fishing		Catch quota	Reduces the risk of over-fishing	Difficult to accurately monitor populations/estimate MSYs/to police maybe unfair for smaller vessels	
		Minimum catch size	Allows fish to reach maturity, Increases chance of breeding and re-population	Some species reach breeding age before maximum size Over-sized fish may be dead by the time they get returned to the sea	
		Net/hook design: Mesh design/size/escape panels Acoustic deterrent devices/dolphin pingers Hook shape/sinkers	Allows immature fish/non-target species to escape Named eg TEDs turtles Reduces risk of bycatch Named eg sinkers Albatross	Expensive	
		Restricted fishing methods Ban on drift nets/ demersal trawling	Avoids bycatch Prevents seabed damage/ habitat destruction	Pressure on other areas	
		Restricted fishing effort Size of boat/engine/time at sea	Reduces risk of over-fishing	May cost in government subsidies to decommission boats	
		No-take zones protected breeding areas closed seasons	Allows breeding and repopulation Conserves other marine biodiversity too	Difficulty of surveillance, enforcement and monitoring of vast offshore area	
		Protected individuals	Ensures breeding and repopulation Named eg lobsters	Pressure on other areas	
		Captive rearing and release to boost wild populations	Increases chance of survival/population/potential catch	Energy use at breeding centre Potential of disease introduction	

3.5.2.2 Fishing	Biodegradable/radio tracked equipment to reduce ghost fishing	Lost nets can be found/broken down reduce ghost fishing accountability	Biodegradable nets may not last very long/be too expensive
3.5.2.3 Aquaculture	Aquaculture	Reduced fishing impacts	Habitat loss Organic waste pollution
		Total =	25

Level	Marks	Descriptor
5	21–25	A comprehensive response with a clear and sustained focus. Content is accurate and detailed. Relationships are identified, reflecting the holistic nature of environmental science and the answer as a whole is coherent. A wide range of relevant natural processes/systems and environmental issues are described and articulated clearly. These are applied systematically to the question, with clear relevance to the context. Where conclusions are made, these are fully supported by judgements and presented in a logical and coherent way. Relevant environmental terminology is used consistently and accurately throughout. If there are errors, these are very minor indeed and not sufficient to detract from the answer.
4	16–20	A response in which the focus is largely sustained, with content that is mainly accurate and detailed. Relationships are identified and the answer is largely coherent. A range of natural processes/systems and environmental issues are described and articulated clearly. In most cases, these are applied appropriately to the question but, in some, it is less clear why they are relevant. Where conclusions are made, these are supported by judgements which are mostly coherent and relevant. Relevant environmental terminology is used consistently and throughout, with no more than minor errors.
3	11–15	A partial response which is focused in parts. The content is mostly accurate but not always detailed. There is an attempt at identifying relationships, but the answer as a whole is not fully coherent. A range of natural processes/systems and environmental issues are described, most are articulated clearly. In some cases, these are applied appropriately to the context but, in most, it is less clear why they are relevant. Where conclusions are made, it is not always clear how they relate to the judgments given and are likely to contain errors. Relevant environmental terminology is used, but not consistently and there may be errors.
2	6–10	An unbalanced response, lacking in focus. The content may be inaccurate and lacking detail. There is some attempt at identifying relationships, but the answer is not coherent. A limited range of natural processes/systems and environmental issues are described but not articulated clearly and likely to contain errors and/or omissions. There is a limited attempt to apply them to the context. Any conclusions are likely to be asserted, with no supporting judgements and fundamental errors. Environmental terminology is used, but not always appropriately and sometimes
1	1–5	Fragmented points, whose relevance to the question and relationships to each other are unclear. A few natural processes/systems and environmental issues are listed, but unlikely to be described and many may be irrelevant. There is no clear attempt to apply them to the context. It is unlikely that a conclusion will be present. There is an attempt to use environmental terminology, but seldom appropriately.
	0	Nothing written worthy of credit.